Thevarajah Kopigan

Colombo, Sri Lanka thevarajahkopigan@gmail.com +94 76 080 4086
Portfolio

Profile

Computer Engineering undergraduate at the University of Sri Jayewardenepura, expected to graduate in 2028. Interested in IoT-based systems, computer vision, robotics, and software development. Academic project experience includes an IoT-based smart farm management system, a computer vision-based power bank detection project, a clinic management system, and a smart maze-following robot.

Education

University of Sri Jayewardenepura **Sri Lanka**
B.Sc. (Hons) in Computer Engineering *Expected Graduation: 2028*

Vavuniya Tamil Madya Maha Vidyalayam **Sri Lanka**
G.C.E. Advanced Level – Physical Science Stream
Combined Mathematics – A Physics – B Chemistry – A

Academic & Technical Projects

- **Smart Farm Management System** *IoT Project*
  - Worked on an IoT-based farm management system using an ESP32 and environmental and soil sensors.
  - Integrated temperature and humidity sensing, soil moisture monitoring, and an NPK sensor for farm-condition monitoring.
  - Included relay-based control for farm equipment such as a water pump.
  - **Technologies/Components:** ESP32, DHT11/DHT22, soil moisture sensor, NPK sensor, relays, water pump.
- **Power Bank Detection System** *Computer Vision Project*
  - Project focused on detecting power banks using computer vision techniques.
  - Specific implementation details and the model/framework can be added once confirmed.
- **Clinic Management System** *Software Development Project*
  - Worked on a clinic management system project.
  - Specific features, technologies, and implementation details can be added once confirmed.
- **Smart Maze-Following Robot** *Robotics Project*
  - Worked on a robot designed to navigate a maze using distance sensing and embedded hardware.
  - Project components include an ESP32, VL53L0X time-of-flight distance sensors, and an MPU6050 inertial sensor.
  - **Technologies/Components:** ESP32, VL53L0X, MPU6050, motors, and motor driver.

Technical Skills & Interests

- **Areas of Interest:** Internet of Things (IoT), embedded systems, computer vision, robotics, software development.
- **Hardware / Platforms:** ESP32, sensor integration, relay-based control.

Languages

- **Tamil:** Native
- **English:** Speaking, reading, and writing
- **Sinhala:** Speaking, reading, and writing